

Hence, if $|z| > (1/A)$, then for an infinity of subscripts n we have $|a_n z^n| > 1$. In this case, the series diverges because we cannot have $\lim_{n \rightarrow \infty} a_n z^n = 0$. If, on the other hand, $|z| < (1/A)$, then there exists an index n_0 such that, for $n > n_0$, $\sqrt[n]{|a_n z^n|} < t$, where $A|z| < t < 1$. For $n > n_0$, the terms of (16) are therefore smaller in absolute value than the corresponding terms of the converging geometric series $\sum_{n=0}^{\infty} t^n$ ($0 < t < 1$). Hence, the series (16) converges.

EXERCISES

1. Find the radii of convergence of the power series $\sum_{n=0}^{\infty} a_n z^n$, where

$$(a) \ a_n = \frac{1}{n^n}.$$

$$(b) \ a_n = \left(1 - \frac{1}{n}\right)^{n^2}.$$

$$(c) \ \frac{1}{n^n}.$$

2. If the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ and $g(z) = \sum_{n=0}^{\infty} b_n z^n$ both converge for $|z| < R$, show that the series $h(z) = \sum_{n=0}^{\infty} c_n z^n$, where $c_n = \sum_{\nu=0}^n a_\nu b_{n-\nu}$, also converges for $|z| < R$ and that $h(z) = f(z)g(z)$.

3. If $f(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!}$, show with the help of the result of the preceding exercise that

$$[f(z)]^2 = f(2z).$$

4. Find a power series which satisfies the differential equation $f'(z) = zf(z)$ and show that it converges for all finite values of z .

5. Find the most general power series (involving two arbitrary constants) which satisfies the differential equation $f''(z) + f(z) = 0$.

6. Show that the value of a power series cannot be constant in the interior of its circle of convergence unless it reduces to its first term.

7. Using the result of the preceding exercise, obtain the power series expansion of the function $f(z) = (1-z)^{-2}$ by means of a recursion formula derived from the identity $(1-z)f'(z) = 1$. Find the radius of convergence of the series in question.

7. Preliminary Study of Some Elementary Functions. As already pointed out in the preceding section, the Taylor series expansions known from the elementary calculus can be made the starting point for the definition of many analytic functions. In the present section we shall discuss the properties of some of these functions. We begin with the study of the exponential function which is defined by the power series

$$e(z) = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \cdots = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

Since, by Cauchy's test, this series converges for all finite values of z , it follows that the analytic function $e(z)$ is regular at all finite points of the complex z -plane. By term-by-term differentiation it is seen that the function $e(z)$ satisfies the differential equation $e'(z) = e(z)$. If a is a constant complex number, it is immediately verified that the derivative of the analytic function $e(z)e(a-z)$ is zero. Hence, this function is necessarily a constant, whose value is found to be $e(a)$ by setting $z = 0$. Thus, $e(z)e(a-z) = e(a)$ or, setting $a - z = w$,

$$e(z)e(w) = e(z+w).$$

The addition theorem of the real function e^z —to which $e(z)$ reduces if z is real—is therefore also verified for the analytic function $e(z)$. With the notation $e(1) = \sum_{n=0}^{\infty} \frac{1}{n!} = e$, it follows from the addition theorem that

$e(2) = e^2$, $e(3) = e^3$, $\dots$, $e(n) = e^n$. Because of this property of the function $e(z)$, we shall henceforth write e^z instead of $e(z)$. The reader will, however, do well to remember that this is a purely symbolical notation; by no stretch of the imagination can the number $e^{1+i} = \sum_{n=0}^{\infty} \frac{(1+i)^n}{n!}$

be regarded as the “ $(1+i)$ th power” of e . An immediate consequence of the addition theorem of e^z is the fact that the exponential function does not take the value zero at any finite point of the complex plane. Indeed, we have

$$e^* e^{-z} = e^0 = 1;$$

since the value of e^{-z} is a well-determined finite number, it follows that e^z cannot be zero as otherwise its product with e^{-z} would also be zero.

Consider now the function

$$e^{iz} = 1 + \frac{(iz)}{1!} + \frac{(iz)^2}{2!} + \frac{(iz)^3}{3!} + \cdots$$

In view of $i^2 = -1$, this may also be written

$$(20) \quad e^{iz} = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots + i \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots \right).$$

The two power series appearing here are formally identical with the well-known Taylor expansions of the real trigonometric functions cosine and sine. The analytic functions which are obtained if z is regarded as a complex variable will also be denoted by the symbols $\cos z$ and $\sin z$, although it goes without saying that, for complex values of z , the trigonometric associations of these functions lose their meaning. The analytic functions $\cos z$ and $\sin z$ are thus defined by the power series

$$\begin{aligned}\cos z &= 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots, \\ \sin z &= z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots.\end{aligned}$$

Since both these series clearly converge for all finite values of z , $\cos z$ and $\sin z$ are regular at all finitely distant points of the z -plane. It is immediately confirmed that

$$\cos(-z) = \cos z, \quad \sin(-z) = -\sin z,$$

(21) that is, $\cos z$ is an *even function* of z and $\sin z$ is an *odd function* of z . It is further apparent from the expansions for $\cos z$ and $\sin z$ that these functions have the differentiation formulas

$$\frac{d}{dz}(\cos z) = -\sin z, \quad \frac{d}{dz}(\sin z) = \cos z.$$

(20) shows that $\cos z$ and $\sin z$ are related to the exponential function by the identity

$$e^{iz} = \cos z + i \sin z.$$

(22) Substituting $-z$ instead of z in (22) and using (21), we obtain

$$e^{-iz} = \cos z - i \sin z.$$

(23) Multiplication of (22) and (23) yields

$$\cos^2 z + \sin^2 z = 1$$

(24) which shows that the relation which, in the trigonometric interpretation of the real functions sine and cosine, expresses the theorem of Pythagoras, remains true if the definition of these functions is extended to complex values of the variable. (22) and (23) can also be used to give alternative definitions of the functions $\cos z$ and $\sin z$. We evidently have

$$\begin{aligned}\cos z &= \frac{1}{2}(e^{iz} + e^{-iz}), & \sin z &= \frac{1}{2i}(e^{iz} - e^{-iz}).\end{aligned}$$

With the help of (22) and the addition theorem, the function e^z can be decomposed into its real and imaginary parts. We obtain

$$e^z = e^{x+iy} = e^x e^{iy} = e^x(\cos y + i \sin y),$$

that is,

$$(26) \quad e^z = e^x \cos y + ie^x \sin y.$$

The absolute value of e^z is therefore

$$|e^z| = e^x(\cos^2 y + \sin^2 y)^{\frac{1}{2}} = e^x.$$

Since the real function e^x assumes the value 1 only for $x = 0$ (e^x increases monotonically with x), the absolute value of e^z will be 1 if, and only if, $x = \operatorname{Re} \{z\} = 0$, that is, if z is pure imaginary. We further observe that, with the help of (22), the polar representation (3), Sec. 2, of a complex number can be cast into a more compact form. If the absolute value r and the argument θ ($r \geq 0$, $0 \leq \theta < 2\pi$) of the complex number $z = x + iy$ are defined by

$$x = r \cos \theta, \quad y = r \sin \theta,$$

then, by virtue of (22),

$$(27) \quad z = re^{i\theta}.$$

The real trigonometric functions $\cos y$ and $\sin y$ are known to have the period 2π , that is, they satisfy the relations $\cos(y + 2\pi) = \cos y$ and $\sin(y + 2\pi) = \sin y$. As shown by (26), the function e^z will therefore also remain unchanged if 2π is added to the imaginary part y of the variable z . Hence, the function e^z has the period $2\pi i$, that is,

$$(28) \quad e^{z+2\pi i} = e^z.$$

This periodicity of e^z can also be derived in a direct fashion which, at the same time, shows that e^z has no periods which are not integral multiples of $2\pi i$. By the addition theorem, we have

$$e^{z+p} = e^z e^p.$$

p will therefore be a period of e^z if, and only if,

$$(29) \quad e^p = 1.$$

As shown above, the absolute value of e^z is equal to 1 only if z is pure imaginary. Hence, $p = it$, where t is real. It thus follows from (22) that we must have $\cos t + i \sin t = 1$ which, of course, is only possible if $\cos t = 1$, $\sin t = 0$. It is known from elementary trigonometry that this occurs for $t = 2\pi$ and for no other value between 0 and 2π . $2\pi i$ and its integral multiples are therefore periods of e^z . If there were a period

$p = it$ which is not of the form $2\pi im$, then there would exist an integer m such that $2\pi m < t < 2\pi(m+1)$, and we would have

$$e^{p+i(t-2\pi m)} = e^p e^{it} e^{-2\pi im} = e^p.$$

The number $i(t - 2\pi m)$, where $0 < t - 2\pi m < 2\pi$, would therefore also be a period, but this was shown to be impossible. Hence, the equation (29) has only solutions of the form $2\pi im$, and these are all the periods of e^z .

The addition theorems of the real trigonometric functions also hold for the analytic functions $\cos z$ and $\sin z$. We thus have

$$\begin{aligned} \cos(z+w) &= \cos z \cos w - \sin z \sin w, \\ \sin(z+w) &= \sin z \cos w + \cos z \sin w. \end{aligned}$$

The proof is an immediate consequence of (25) and (22) and is left as an exercise for the reader.

We now turn to the function $\log z$ which, exactly as in the case of the real logarithm, is defined as the function inverse to the exponential function. The fact that the exponential function and the logarithm are inverse to each other is expressed by saying that the two equations

$$(32) \quad w = \log z, \quad z = e^w$$

are equivalent to each other. We first prove that $\log z$ is indeed an analytic function of z by showing that, for $z \neq 0$, $f(z) = \log z$ has a derivative. We have, by (32),

$$\begin{aligned} f'(z) &= \lim_{z_1 \rightarrow z} \frac{f(z_1) - f(z)}{z_1 - z} = \lim_{w_1 \rightarrow w} \frac{w_1 - w}{e^{w_1} - e^w} \\ &= \frac{1}{\lim_{w_1 \rightarrow w} \frac{e^{w_1} - e^w}{w_1 - w}} = \frac{1}{\frac{d}{dw}(e^w)} = \frac{1}{e^w}. \end{aligned}$$

In view of $z = e^w$, we finally obtain

$$\frac{d}{dz}(\log z) = \frac{1}{z},$$

which shows that $\log z$ has a derivative, provided $z \neq 0$. The function $\log z$ is therefore regular at all points of the z -plane except at the origin.

The function $\log z$ has a peculiar property which was not displayed by the analytic functions considered so far. It is *many-valued*. In order to understand this property, we recall the periodicity of the exponential function, that is, the identity

$$e^{w+2\pi im} = e^w, \quad m = 0, \pm 1, \pm 2, \dots$$

The inverse of the exponential function defined by (32) may therefore be taken to be any of the functions $\log z + 2\pi im$ ($m = 0, \pm 1, \pm 2, \dots$); in other words, the function $\log z$ is only defined up to an integral multiple of $2\pi i$. This may appear to the beginner as a source of confusion; however, order is readily restored by means of a more penetrating study of the properties of the function $\log z$. Writing $w = u + iv$, we find from (32) and (26) that

$$z = x + iy = e^u(\cos v + i \sin v),$$

whence $|z| = e^u$ and $(y/x) = \tan v$. If "log" now denotes the real logarithm of a positive number, as known from real analysis (it is somewhat inconsistent to use this notation for both this real function and the many-valued analytic function $\log z$; however, confusion can hardly arise because of this practice), we have $\log(e^u) = u$, whence

$$u = \log |z| = \log r, \quad z = re^{i\theta}.$$

In view of $(y/x) = \tan v$, we have $v = \tan^{-1}(y/x) = \theta$, where the angle θ is only determined up to an integral multiple of 2π . With the expressions for $u = \operatorname{Re} \{\log z\}$ and $v = \operatorname{Im} \{\log z\}$ thus obtained, we have

$$(33) \quad \log z = \log r + i\theta, \quad z = re^{i\theta}.$$

The indeterminacy of the logarithm is therefore simply an expression of the fact that the polar coordinates of a point in the plane are not uniquely determined; we can always add an integral multiple of 2π to the angle θ without changing the position of the point which is described by the polar coordinates. In analytic geometry, this many-valuedness of the polar coordinate system is decreed out of existence by the simple expedient of prescribing that the angle θ is always to be taken between 0 and 2π . We may employ the same device in order to turn the function $\log z$ into a single-valued analytic function; however, it is customary to restrict the value of θ to the interval $-\pi < \theta \leq \pi$, rather than to the interval $0 \leq \theta < 2\pi$. The value of the function $\log z$ obtained by this restriction is called the *principal value* of $\log z$. It is defined by (33), where $\log r$ denotes the real logarithm of the positive number r , with the additional condition that $-\pi < \theta \leq \pi$. In order to understand the connection between the principal value of $\log z$ and its other values, namely, those obtained by adding $2\pi m$ to θ , we consider a point $z = re^{i\theta}$ and let this point traverse the circle $|z| = r$ until it returns to its original position. In this operation, the angle θ will have increased or decreased by 2π , according as the point traveled in the positive or negative direction. (33) shows that, at the same time, the value of $\log z$ was changed by an amount

of $\pm 2\pi i$, the sign depending on whether the origin was surrounded in the positive or negative direction. If the point z surrounds the origin m times (where m may be positive, negative, or zero), we shall therefore arrive at the value

$$(34) \quad \log z = \log r + i\theta + 2\pi im, \quad z = re^{i\theta}, \quad -\pi < \theta \leq \pi.$$

This situation may also be described by saying that the function $\log z$ has an infinity of branches all of which are single-valued; in order to get from one branch to another in a continuous manner, we have to surround the origin a suitable number of times.

The condition $-\pi < \theta \leq \pi$ which determined the single-valued principal value of $\log z$ can be restated in a more graphic fashion. Since both $\theta = \pi$ and $\theta = -\pi$ can be regarded as the equation of the negative axis in the z -plane, the condition $-\pi < \theta \leq \pi$ "forbids" the passing of the point z across the negative axis. The function $\log z$ will therefore be a single-valued function of z if a "cut" along the whole negative axis is applied to the z -plane: In accordance with the restriction $-\pi < \theta \leq \pi$, the "upper edge" of the cut belongs to the cut plane and the "lower edge" does not. The placing of the cut along the negative axis is, of course, a pure matter of convention; the reader will easily verify that any curve that does not intersect itself and connects the origin with the point at infinity will serve just as well. In the cut plane, not only the principal value but also all the other values of the function $\log z$ are single-valued functions. The desire to define a region in which the function $\log z$ in its entirety is single-valued may therefore suggest the following device: To each branch of the function $\log z$ we assign one replica of the cut plane in which this branch is single-valued; these cut planes can be ordered according to the integer m ($m = \dots, -1, 0, 1, \dots$) in (34) by which they are characterized. We now stack all these cut planes upon each other in such a fashion that the $(m+1)$ st plane lies immediately on top of the m th plane and that the corresponding points in all planes lie exactly on top of each other. In order to take account of the fact that we pass from the m th branch of $\log z$ to the $(m+1)$ st branch if the point z surrounds the origin in the positive direction, we now connect the upper edge of the cut in the m th plane with the lower edge of the cut in the $(m+1)$ st plane. If these connections are effected throughout the whole infinite set of planes (m runs from $-\infty$ to $+\infty$), we arrive at what is known as the *Riemann surface of the function* $\log z$. On this Riemann surface, $\log z$ is a single-valued function. It is interesting to note that this surface has only two boundary points, namely, the points $z = 0$ and $z = \infty$. All the other possible boundary points satisfy $0 > z > -\infty$; but, since each sheet of the Riemann surface is connected with its two adjacent sheets along the two

edges of the cut $0 > z > -\infty$, these points are—in all sheets—situated in the interior of the surface.

We close this chapter with a discussion of the function $f(z) = z^\alpha$, where α is an arbitrary complex number. If α is a positive integer, say n , then z^n is defined as a product of n factors all of which are equal to z . The easiest way of generalizing this definition to the case in which α is an arbitrary complex number is by means of the function $\log z$. We define

$$(35) \quad z^\alpha = e^{\alpha \log z}.$$

The reader will verify without difficulty that if α is an integer or a rational number, this definition coincides with the usual definition of a power of z . Clearly, the many-valuedness of the function $\log z$ will generally result in the many-valuedness of the function z^α . If, in (35), $\log z$ denotes a definite branch, say, the principal value, of the logarithm, the various possible values of z^α will be of the form

$$z^\alpha = e^{\alpha(\log z + 2\pi im)} = e^{\alpha \log z} e^{2\pi i \alpha m},$$

where m is an arbitrary integer. The various different values of z^α will therefore be obtained from any one of them by multiplication with a factor $e^{2\pi i \alpha m}$ ($m = \dots, -1, 0, 1, \dots$). As a result, z^α will in general have an infinity of different values. However, there are cases in which the number of the different values of z^α is finite; obviously, these correspond to the cases in which only a finite number of the values

$$e^{2\pi i \alpha m}, \quad m = \dots, -1, 0, 1, \dots$$

are different from each other. If this occurs, then there must be two integers m and m' ($m' \neq m$) such that $e^{2\pi i \alpha m} = e^{2\pi i \alpha m'}$ or, in view of the addition theorem of the exponential function,

$$e^{2\pi i \alpha (m-m')} = 1.$$

Since e^x takes the value 1 only if $x = 2\pi in$, where n is an integer, this is equivalent to $\alpha(m-m') = n$. It follows that α is a rational number. On the other hand, if α is a rational number of the form m/n (m and n having no common factors and $n \geq 1$), then the set

$$e^{2\pi i \frac{m}{n} \mu}, \quad \mu = 0, \pm 1, \pm 2, \dots,$$

clearly contains only n different numbers; as such we may take those corresponding to $\mu = 0, 1, \dots, n-1$. Hence, the function z^α is n -valued if, and only if, α is a rational number m/n , where m and n have no common factors; if α is not rational, z^α has an infinity of values.

In view of (35), the function z^α is single-valued on the Riemann surface

of the logarithm; each branch can be made single-valued in the z -plane by the introduction of a cut along the negative axis. However, if α is rational, it is possible to construct a Riemann surface which has only a finite number of sheets and on which z^α is single-valued. As an example, consider the function $f(z) = z^{\frac{1}{2}} = \sqrt{z}$. In view of the foregoing, this function is two-valued; it has two branches which, for $z = 1$, reduce to 1 and -1 , respectively. We have

$$\sqrt{z} = \sqrt{r} e^{i\theta/2}, \quad z = r e^{i\theta},$$

where $\sqrt{r}$ is the positive square root of r . If we surround the origin in the positive direction, that is, if θ grows by the amount 2π , $\sqrt{z}$ will assume the value

$$\sqrt{r} e^{i(\theta+2\pi)/2} = \sqrt{r} e^{i\theta/2} e^{i\pi} = -\sqrt{r} e^{i\theta/2} = -\sqrt{z}.$$

Another closed circuit will again result in multiplication with the factor -1 . Hence, two closed circuits about the origin will bring us back to the original value of the function. In order to construct a Riemann surface that takes account of these facts, we proceed as follows: We take two replicas of the z -plane both of which have been furnished with a cut from $z = \infty$ to $z = 0$ along the negative axis and place one on top of the other; to make things definite we suppose that the upper plane corresponds to the branch of $\sqrt{z}$ for which $\sqrt{1} = 1$ and the lower plane to that for which $\sqrt{1} = -1$. We connect the upper edge of the cut in the upper plane with the lower edge of the cut in the lower plane. It should be noted that the two connections of the two sheets penetrate each other and that it is consequently impossible to build a correct paper model of this Riemann surface. Obviously, the function $f(z) = \sqrt{z}$ is two-valued in any domain which contains either of these points and therefore, in view of our definition of regularity, $f(z)$ cannot be regular throughout these domains. Taking, in particular, the domains $|z| < \epsilon$ and $|z| > \frac{1}{\epsilon}$ (ϵ arbitrarily small) which contain the points $z = 0$ and $z = \infty$, respectively, we conclude that $z = 0$ and $z = \infty$ are singular points of the function $f(z) = \sqrt{z}$. A singular point of this type, that is, a point a such that there exists no neighborhood $|z - a| < \epsilon$ in which the function $f(z)$ is single-valued, is called a *branch point* of $f(z)$. If, for arbitrary small ϵ , the function is n -valued in $|z - a| < \epsilon$, then we say that a is a *branch point of order $n - 1$* of $f(z)$. $z = 0$ and $z = \infty$ are thus branch

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points of first order of $f(z) = \sqrt{z}$. The singularities of the function $f(z) = \log z$ at $z = 0$ and $z = \infty$ may be regarded as branch points of infinite order.

The construction of a Riemann surface for the n -valued function $z^{\frac{1}{n}} = \sqrt[n]{z}$ is effected along the same lines as in the case $n = 2$. We take n replicas of the z -plane and cut them along the negative axis. We place them one upon another and connect the edges of the cuts in the following manner: We connect the upper edge of the cuts in the sheet with the lower edge of the second sheet, the upper edge of the second sheet with the lower edge of the third sheet, etc. If this has been done for each pair of adjacent sheets, there remain only two unconnected edges, namely, the lower edge of the first sheet and the upper edge of the last sheet. We then connect these two edges and thus obtain a closed n -sheeted Riemann surface. The proof that the function $f(z) = \sqrt[n]{z}$ is indeed single-valued on this Riemann surface is left as an exercise to the reader.

EXERCISES

1. The hyperbolic functions are defined by $\cosh z = \cos iz$ and $i \sinh z = \sin iz$. Show that the functions are real for real values of z , and use your result and the addition theorems of the trigonometric functions in order to decompose $\sin z$ and $\cos z$ into their real and imaginary parts.
2. Obtain the power series expansion of the function $f(z) = \frac{1+z}{1-z}$ and, by taking real parts and using (27), derive the identity (23') of Sec. 6, Chap. I.
3. By applying a similar procedure to the series obtained in Exercise 7 of the preceding section, obtain closed expressions for $\sum_{n=1}^{\infty} n r^n \cos n\theta$ and $\sum_{n=1}^{\infty} n r^n \sin n\theta$ ($0 < r < 1$).
4. Prove the periodicity of $\sin z$ and $\cos z$ from their addition theorems, the identity (24), and the fact that $\sin \frac{1}{2}\pi = 1$.
5. Use (22) and (25) to prove the trigonometric identities
 - (a) $\frac{1}{2} + \cos \theta + \cos 2\theta + \cdots + \cos n\theta = \frac{\sin \frac{1}{2}(2n+1)\theta}{2 \sin \frac{1}{2}\theta}$,
 - (b) $\cos \theta + \cos 3\theta + \cdots + \cos (2n-1)\theta = \frac{\sin 2n\theta}{2 \sin \theta}$,
 - (c) $\sin \theta + \sin 2\theta + \cdots + \sin n\theta = \frac{\sin \frac{1}{2}\theta \sin \frac{1}{2}(n+1)\theta}{\sin \frac{1}{2}\theta}$.
6. If t is a real parameter, growing from $-\infty$ to ∞ , and a, b are constants, show that the points $z = \cos(a+it)$ and $z = \cos(t+ib)$ describe, respectively, hyperbolas and ellipses whose foci are at $z = \pm 1$.
7. Prove the addition theorem of the function e^z by direct multiplication of the series for e^z and e^w , and suitable rearrangement of terms.
8. Solving the differential equation $f''(z) + f(z) = 0$ by means of a power series with indeterminate coefficients, show that the most general analytic solution of this equation can be obtained in either of the forms $A \cos z + B \sin z$ or $C e^{iz} + D e^{-iz}$,